

Jeon, Sanghun

professor

基本信息

姓名: Jeon, Sanghun

所属院校: 韩国科学技术院

邮箱: jeonsh@kaist.ac.kr

地区: South Korea

年龄范围: 50岁以上

学科领域

材料物理

材料科学与工程

功能材料

集成电路设计与集成系统

微电子科学与工程

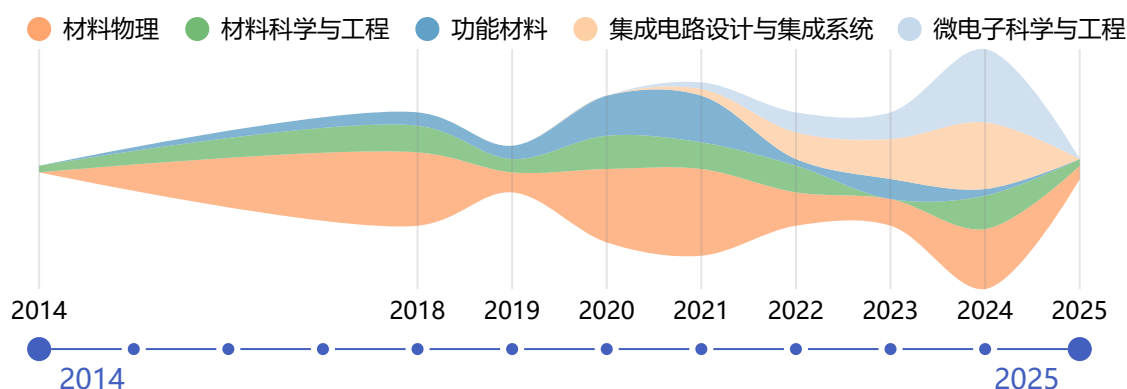
研究领域

薄膜材料与表面物理

个人简介

Jeon, Sanghun is a professor at KAIST. He holds a Ph.D. from GIST in 2003. His research areas include Human Computer Interaction, Integrated Circuits & Semiconductor Systems, etc. Currently, he is researching core materials, process, device and integration technology of wearable IoT sensor, hafnia ferroelectric device, etc. He leads the ANTONIS Lab at the National Nano Fab Center (E19), S - 217. His extension is #7444, and his email is jeonsh@kaist.ac.kr. His website is <http://antonis.kaist.ac.kr>. He has notable achievements in top - tier journals like Nature Materials and Advanced Materials.

近年发表的论文



Enhancement of performance and reliability in MOSFETs via high-pressure microwave annealing

JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B

Kang, Geonhyeong; Shin, Hunbeom; Kim, Seungyeob; Zhang, Lingwei; Kim, Giuk; Lee, Sujeong; Nam, Yunseok; Kim, Chaeheon; Kim, Hoon; Ahn, Jinho; Jeon, Sanghun

被引用次数: 0

2025

Very high frequency (~100 MHz) plasma enhanced atomic layer deposition high-κ hafnium zirconium oxide capacitors near morphotropic phase boundary with low current density & high-κ for DRAM technology

JOURNAL OF MATERIALS CHEMISTRY C

Yang, Ketong; Shin, Hunbeom; Kim, Seungyeob; Jung, Taeseung; Jeon, Sanghun

被引用次数: 0

2025

Monolithic three-dimensional hafnia-based artificial nerve system

NANO ENERGY

Jung, Minhyun; Kim, Seungyeob; Hwang, Junghyeon; Kim, Chaeheon; Kim, Hye Jin; Kim, Yun-Jeong; Jeon, Sanghun

被引用次数: 2

2024

Enhancing performance of microbolometers by utilizing low-temperature polycrystalline silicon

JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B

Jung, Taeseung; Kim, Seungyeob; Lee, Sangho; Ahn, Jinho; Jeon, Sanghun

被引用次数: 0

2024

Flexible Artificial Mechanoreceptor Based on Microwave Annealed Morphotropic Phase Boundary of HfxZr1-xO2 Thin Film

ADVANCED ELECTRONIC MATERIALS

Jung, Minhyun; Kim, Seungyeob; Hwang, Junghyeon; Kim, Hye Jin; Kim, Yunjeong; Ahn, Jinho; Jeon, Sanghun

被引用次数: 5

2024

Damage mitigation as a strategy to achieve high ferroelectricity and reliability in hafnia for random-access-memory

JOURNAL OF MATERIALS CHEMISTRY C

Hwang, Junghyeon; Shin, Hunbeom; Kim, Chaeheon; Ahn, Jinho; Jeon, Sanghun

被引用次数: 0

2024

Dual-port ferroelectric NAND flash memory for large memory window, QLC programmable and disturbance-free operations

JOURNAL OF MATERIALS CHEMISTRY C

Joh, Hongrae; Lee, Sangho; Ahn, Jinho; Jeon, Sanghun

被引用次数: 0

2024

The Effect of Ferroelectric/Dielectric Capacitance Ratio on Short-Term Retention Characteristics of MFMIS FeFET

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Hwang, Junghyeon; Kim, Giuk; Joh, Hongrae; Ahn, Jinho; Jeon, Sanghun

被引用次数: 0

2024

Unlocking the Potential of Hafnia Ferroelectrics: Achieving High Reliability via Plasma Frequency Modulation in Very High-Frequency Plasma-Enhanced Atomic Layer Deposition

ACS APPLIED ELECTRONIC MATERIALS

Yang, Ketong; Jung, Minhyun; Jung, Taeseung; Yoon, Jae Seok; Hwang, Junghyeon; Shin, Hunbeom; Kim, Seungyeob; Kim, Chaeheon; Jeon, Sanghun

被引用次数: 0

2024

Monolithically Integrated Complementary Ferroelectric FET XNOR Synapse for the Binary Neural Network

ACS APPLIED MATERIALS & INTERFACES

Hwang, Junghyeon; Joh, Hongrae; Kim, Chaeheon; Ahn, Jinho; Jeon, Sanghun

被引用次数: 6

2024

Physics, Structures, and Applications of Fluorite-Structured Ferroelectric Tunnel Junctions

SMALL

Hwang, Junghyeon; Goh, Youngin; Jeon, Sanghun

被引用次数: 15

2024

Enhanced performance of hafnia self-rectifying ferroelectric tunnel junctions at cryogenic temperatures

NANO CONVERGENCE

Hwang, Junghyeon; Kim, Chaeheon; Ahn, Jinho; Jeon, Sanghun

被引用次数: 0

2024

Design Guidelines of Hafnia Ferroelectrics and Gate-Stack for Multilevel-Cell FeFET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Sangho; Kim, Giuk; Lee, Youngkyu; Shin, Hunbeom; Jeong, Youngseok; Zhang, Lingwei; Jung, Seong-Ook; Jeon, Sanghun

被引用次数: 2

2024

Novel strategies for low-voltage NAND flash memory with negative capacitance effect

JAPANESE JOURNAL OF APPLIED PHYSICS

Kim, Giuk; Kim, Taeho; Lee, Sangho; Hwang, Junghyeon; Jung, Minhyun; Ahn, Jinho; Jeon, Sanghun

被引用次数: 1

2024

Positive Interaction Between Charge Trapping and Polarization Switching in Metal-Interlayer-Ferroelectric-Interlayer-Silicon (MIFIS) Ferroelectric Field-Effect Transistor

IEEE ELECTRON DEVICE LETTERS

Choi, Hyojun; Kim, Giuk; Lee, Sangho; Shin, Hunbeom; Lim, Youngjin; Kim, Kang; Kim, Do-Hyung; Oh, Il-Kwon; Park, Sang-Hee Ko; Ahn, Jinho; Jeon, Sanghun

被引用次数: 0

2024

Optimizing De-Trap Pulses in Gate-Injection Type Ferroelectric NAND Cells to Minimize Read After Write Delay Issue

IEEE ELECTRON DEVICE LETTERS

Kim, Giuk; Choi, Hyojun; Cho, Hongrae; Lee, Sangho; Shin, Hunbeom; Kang, Hyunjun; Kim, Hoon; Shin, Seokjoong; Park, Seonjae; Kwon, Sunseong; Lim, Youngjin; Kim, Kang; Chung, Jong Min; Oh, Il-Kwon; Park, Sang-Hee Ko; Ahn, Jinho; Jeon, Sanghun

被引用次数: 0

2024

Ultrafast (50 ns) ID-VG Analysis on Oxide Thin-Film Transistors With Morphotropic Phase Boundary State High- κ Gate Insulator

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jung, Taeseung; Nam, Sooji; Jeon, Sanghun

被引用次数: 0

2024

Stabilization of Morphotropic Phase Boundary in Hafnia via Microwave Low-Temperature Crystallization Process for Next-Generation Dynamic Random Access Memory Technology

PHYSICA STATUS SOLIDI-RAPID RESEARCH LETTERS

Shin, Hunbeom; Kim, Giuk; Lee, Sujeong; Choi, Hyojun; Lee, Sangho; Lee, Sangmok; Nam, Yunseok; Kang, Geonhyeong; Kim, Hyungjun; Ahn, Jinho; Jeon, Sanghun

被引用次数: 0

2024

HZO (> 10 nm) Films for Achieving High-TEXPRESERVE1 Near Morphotropic Phase Boundary at Low-Temperature Furnace Annealing Process

IEEE TRANSACTIONS ON ELECTRON DEVICES

Qin, Yixin; Gaddam, Venkateswarlu; Jung, Taeseung; Jeon, Sanghun

被引用次数: 0

2024

Novel Strategies Toward High-performance FeFET for Computing In Memory Application

2024 INTERNATIONAL VLSI SYMPOSIUM ON TECHNOLOGY, SYSTEMS AND APPLICATIONS, VLSI TSA

Kim, Giuk; Choi, Hyojun; Lee, Sangho; Shin, Hunbeom; Zhang, Lingwei; Lee, Sangmok; Nam, Yunseok; Kim, Woongjin; Ock, Jihye; Lee, Sujeong; Kang, Hyunjun; Jeon, Sanghun

被引用次数: 0

2024

Charge Trap Flash with Superior Program Efficiency by Negative Capacitance-boosting Effect

2024 INTERNATIONAL VLSI SYMPOSIUM ON TECHNOLOGY, SYSTEMS AND APPLICATIONS, VLSI TSA

Kim, Giuk; Choi, Hyojun; Lee, Sangho; Shin, Hunbeom; Zhang, Lingwei; Lee, Sangmok; Nam, Yunseok; Kim, Woongjin; Ock, Jihye; Lee, Sujeong; Kang, Hyunjun; Jeon, Sanghun

被引用次数: 0

2024

Design of Physically Unclonable Function Using Ferroelectric FET With Auto Write-Back Technique for Resource-Limited IoT Security

IEEE INTERNET OF THINGS JOURNAL

Lim, Sehee; Hwang, Junghyeon; Ko, Dong Han; Kim, Se Keon; Oh, Tae Woo; Jeon, Sanghun; Jung, Seong-Ook

被引用次数: 0

2024

Experimental Analysis on the Interaction Between Interface Trap Charges and Polarization on the Memory Window of Metal-Ferroelectric-Insulator-Si (MFIS) FeFET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Giuk; Choi, Hyojun; Lee, Sangho; Shin, Hunbeom; Lee, Sangmok; Nam, Yunseok; Kang, Hyunjun; Shin, Seokjoong; Kim, Hoon; Lim, Youngjin; Kim, Kang; Oh, Il-Kwon; Park, Sang-Hee Ko; Ahn, Jinho; Jeon, Sanghun

被引用次数: 0

2024

First Demonstration of Thermally Stable Zr:HfO₂ Ferroelectrics via Inserting AlN Interlayer

IEEE ELECTRON DEVICE LETTERS

Lee, Sangmok; Kim, Giuk; Lee, Sangho; Shin, Hunbeom; Lim, Youngjin; Kim, Kang; Kim, Do-Hyung; Oh, Il-Kwon; Ko Park, Sang-Hee; Ahn, Jinho; Jeon, Sanghun

被引用次数: 0

2024

Kinetically Stabilized Hafnia Ferroelectric of Al-Doped HfO₂ Film by Fast Ramping and Fast Cooling Process

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zhang, Lingwei; Kim, Giuk; Lee, Sangho; Shin, Hunbeom; Lim, Youngjin; Kim, Kang; Oh, Il-Kwon; Park, Sang-Hee Ko; Ahn, Jinho; Jeon, Sanghun

被引用次数: 0

2024

Enhancement of electrical properties of a-IGZO thin film transistor by low temperature (150 °C) microwave annealing for flexible electronics

JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B

Jung, Taeseung; Han, Jung Hoon; Nam, Sooji; Jeon, Sanghun

被引用次数: 2

2023

Effect of Floating Gate Insertion on the Analog States of Ferroelectric Field-Effect Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Sangho; Lee, Youngkyu; Kim, Giuk; Kim, Taeho; Eom, Taehyong; Jung, Seong-Ook; Jeon, Sanghun

被引用次数: 8

2023

Oxygen Vacancy Control as a Strategy to Enhance Imprinting Effect in Hafnia Ferroelectric Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jeong, Yeongseok; Gaddam, Venkateswarlu; Goh, Youngin; Shin, Hunbeom; Lee, Sangho; Kim, Giuk; Jeon, Sanghun

被引用次数: 8

2023

High Pressure Microwave Annealing Effect on Electrical Properties of Hf x Zr_{1-x} O Films near Morphotropic Phase Boundary

ACS APPLIED ELECTRONIC MATERIALS

Jung, Minhyun; Kim, Chaeheon; Hwang, Junghyeon; Kim, Giuk; Shin, Hunbeom; Gaddam, Venkateswarlu; Jeon, Sanghun

被引用次数: 3

2023

Dual-Mode Operations of Self-Rectifying Ferroelectric Tunnel Junction Crosspoint Array for High-Density Integration of IoT Devices

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lim, Sehee; Goh, Youngin; Lee, Young Kyu; Ko, Dong Han; Hwang, Junghyeon; Jeong, Yeongseok; Shin, Hunbeom; Jeon, Sanghun; Jung, Seong-Ook

被引用次数: 9

2023

The Opportunity of Negative Capacitance Behavior in Flash Memory for High-Density and Energy-Efficient In-Memory Computing Applications

ADVANCED FUNCTIONAL MATERIALS

Kim, Taeho; Kim, Giuk; Lee, Young Kyu; Ko, Dong Han; Hwang, Junghyeon; Lee, Sangho; Shin, Hunbeom; Jeong, Yeongseok; Jung, Seong-Ook; Jeon, Sanghun

被引用次数: 6

2023

Design consideration of ferroelectric field-effect-transistors with metal-ferroelectric-metal capacitor for ternary content addressable memory

SOLID-STATE ELECTRONICS

Yi, Boram; Hwang, Junghyeon; Oh, Tae Woo; Jeon, Sanghun; Jung, Seong-Ook; Yang, Ji-Woon

被引用次数: 1

2023

Ferroelectric Hafnia-Based M3D FeTFTs Annealed at Extremely Low Temperatures and TCAM Cells for Computing-in-Memory Applications

ACS APPLIED MATERIALS & INTERFACES

Joh, Hongrae; Nam, Sooji; Jung, Minhyun; Shin, Hunbeom; Cho, Sung Haeng; Jeon, Sanghun

被引用次数: 2

2023

A method of controlling the imprint effect in hafnia ferroelectric device

APPLIED PHYSICS LETTERS

Shin, Hunbeom; Gaddam, Venkateswarlu; Goh, Youngin; Jeong, Yeongseok; Kim, Giuk; Qin, Yixin; Jeon, Sanghun

被引用次数: 9

2023

Power-Delay Area-Efficient Processing-In-Memory Based on Nanocrystalline Hafnia Ferroelectric Field-Effect Transistors

ACS APPLIED MATERIALS & INTERFACES

Kim, Giuk; Ko, Dong Han; Kim, Taeho; Lee, Sangho; Jung, Minhyun; Lee, Young Kyu; Lim, Sehee; Jo, Minyoung; Eom, Taehyong; Shin, Hunbeom; Jeong, Yeongseok; Jung, Seongook; Jeon, Sanghun

被引用次数: 13

2023

Flexible and stretchable conductive fabric for temperature detection

2022 IEEE INTERNATIONAL CONFERENCE ON FLEXIBLE AND PRINTABLE SENSORS AND SYSTEMS (IEEE FLEPS 2022)

Eom, Taehyong; Jung, Minhyun; Bae, Jihyun; Jeon, Sanghun

被引用次数: 0

2022

Vertical-Pillar Ferroelectric Field-Effect-Transistor Memory

PHYSICA STATUS SOLIDI-RAPID RESEARCH LETTERS

Lee, Sangho; Kim, Giuk; Kim, Taeho; Eom, Taehyong; Jeon, Sanghun

被引用次数: 5

2022

Relatively Low-k Ferroelectric Nonvolatile Memory Using Fast Ramping Fast Cooling Annealing Process

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hwang, Junghyeon; Kim, Minki; Jung, Minhyun; Kim, Taeho; Goh, Youngin; Lee, Yongsun; Jeon, Sanghun

被引用次数: 11

2022

Sub 5 Å-EOT Hf_xZr_{1-x}O₂ for Next-Generation DRAM Capacitors Using Morphotropic Phase Boundary and High-Pressure (200 atm) Annealing With Rapid Cooling Process

IEEE TRANSACTIONS ON ELECTRON DEVICES

Das, Dipjyoti; Buyantogtokh, Batzorig; Gaddam, Venkateswarlu; Jeon, Sanghun

被引用次数: 22

2022

Design Guidelines of Thermally Stable Hafnia Ferroelectrics for the Fabrication of 3D Memory Devices

2022 INTERNATIONAL ELECTRON DEVICES MEETING, IEDM

Kim, Giuk; Shin, Hunbeom; Eom, Taehyong; Jung, Minhyun; Kim, Taeho; Lee, Sangho; Kim, Minki; Jeong, Yeongseok; Kim, Ji-Sung; Nam, Kab-Jin; Kuh, Bong Jin; Jeon, Sanghun

被引用次数: 6

2022

Wearable pressure sensor based on solution-coated fabric for pulse detection

2022 IEEE INTERNATIONAL CONFERENCE ON FLEXIBLE AND PRINTABLE SENSORS AND SYSTEMS (IEEE FLEPS 2022)

Eom, Taehyong; Kim, Kyungkwan; Jung, Minhyun; Bae, Jihyun; Jeon, Sanghun

被引用次数: 0

2022

Steep-Slope Transistor with an Imprinted Antiferroelectric Film

ACS APPLIED MATERIALS & INTERFACES

Lee, Sangho; Lee, Yongsun; Kim, Taeho; Kim, Giuk; Eom, Taehyong; Shin, Hunbeom; Jeong, Yeongseok; Jeon, Sanghun

被引用次数: 4

2022

Novel Approach to High κ (~ 59) and Low EOT (~ 3.8 Å) near the Morphotropic Phase Boundary with AFE/FE (ZrO₂/HZO) Bilayer Heterostructures and High-Pressure Annealing

ACS APPLIED MATERIALS & INTERFACES

Gaddam, Venkateswarlu; Kim, Giuk; Kim, Taeho; Jung, Minhyun; Kim, Chaeheon; Jeon, Sanghun

被引用次数: 21

2022

High performance ferroelectric field-effect transistors for large memory-window, high-reliability, high-speed 3D vertical NAND flash memory

JOURNAL OF MATERIALS CHEMISTRY C

Kim, Giuk; Lee, Sangho; Eom, Taehyong; Kim, Taeho; Jung, Minhyun; Shin, Hunbeom; Jeong, Yeongseok; Kang, Myounggon; Jeon, Sanghun

被引用次数: 22

2022

Flexible Ferroelectric Hafnia-Based Synaptic Transistor by Focused-Microwave Annealing

ACS APPLIED MATERIALS & INTERFACES

Joh, Hongrae; Jung, Minhyun; Hwang, Junghyeon; Goh, Youngin; Jung, Taeseung; Jeon, Sanghun

被引用次数: 37

2022

Effects of iCVD organic passivation in oxide thin-film transistors under repetitive bending stress for electrical and mechanical stability

JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B

Jung, Taeseung; Jeon, Sanghun

被引用次数: 2

2022

Non-Volatile Majority Function Logic Using Ferroelectric Memory for Logic in Memory Technology

IEEE ELECTRON DEVICE LETTERS

Hwang, Junghyeon; Lim, Sehee; Kim, Giuk; Jung, Seong-Ook; Jeon, Sanghun

被引用次数: 10

2022

A review on morphotropic phase boundary in fluorite-structure hafnia towards DRAM technology

NANO CONVERGENCE

Jung, Minhyun; Gaddam, Venkateswarlu; Jeon, Sanghun

被引用次数: 39

2022

High Performance and Self-rectifying Hafnia-based Ferroelectric Tunnel Junction for Neuromorphic Computing and TCAM Applications

2021 IEEE INTERNATIONAL ELECTRON DEVICES MEETING (IEDM)

Goh, Youngin; Hwang, Junghyeon; Kim, Minki; Jung, Minhyun; Lim, Sehee; Jung, Seong-Ook; Jeon, Sanghun

被引用次数: 31

2021

Ferroelectricity in Al₂O₃/Hf_{0.5}Zr_{0.5}O₂ Bilayer Stack: Role of Dielectric Layer Thickness and Annealing Temperature

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Das, Dipjyoti; Gaddam, Venkateswarlu; Jeon, Sanghun	2021
被引用次数: 10	
Stress Engineering as a Strategy to Achieve High Ferroelectricity in Thick Hafnia Using Interlayer	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Joh, Hongrae; Jung, Taeseung; Jeon, Sanghun	2021
被引用次数: 32	
Influence of High-Pressure Annealing Conditions on Ferroelectric and Interfacial Properties of Zr-Rich HfxZr1-xO2 Capacitors	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Das, Dipjyoti; Buyantogtokh, Batzorig; Gaddam, Venkateswarlu; Jeon, Sanghun	2021
被引用次数: 33	
The Influence of Top and Bottom Metal Electrodes on Ferroelectricity of Hafnia	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Lee, Yongsun; Goh, Youngin; Hwang, Junghyeon; Das, Dipjyoti; Jeon, Sanghun	2021
被引用次数: 115	
Effect of Insertion of Dielectric Layer on the Performance of Hafnia Ferroelectric Devices	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Hwang, Junghyeon; Goh, Youngin; Jeon, Sanghun	2021
被引用次数: 39	
Effect of high pressure anneal on switching dynamics of ferroelectric hafnium zirconium oxide capacitors	
JOURNAL OF APPLIED PHYSICS	
Buyantogtokh, Batzorig; Gaddam, Venkateswarlu; Jeon, Sanghun	2021
被引用次数: 19	
Interfacial Dipole Modulation Device With SiOX Switching Species	
IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY	
Kim, Giuk; Kim, Taeho; Jeon, Sanghun	2021
被引用次数: 0	
Oxide electronics: Translating materials science from lab-to-fab	
MRS BULLETIN	
Nathan, Arokia; Jeon, Sanghun	2021
被引用次数: 12	
Selector-less Ferroelectric Tunnel Junctions by Stress Engineering and an Imprinting Effect for High-Density Cross-Point Synapse Arrays	
ACS APPLIED MATERIALS & INTERFACES	
Goh, Youngin; Hwang, Junghyeon; Kim, Minki; Lee, Yongsun; Jung, Minhyun; Jeon, Sanghun	2021
被引用次数: 48	

Ferroelectricity Enhancement in Hf_{0.5}Zr_{0.5}O₂ Based Tri-Layer Capacitors at Low-Temperature (350 °C) Annealing Process

IEEE ELECTRON DEVICE LETTERS

Gaddam, Venkateswarlu; Das, Dipjyoti; Jung, Taeseung; Jeon, Sanghun

被引用次数: 44

2021

Enabling large ferroelectricity and excellent reliability for ultra-thin hafnia-based ferroelectrics with a W bottom electrode by inserting a metal-nitride diffusion barrier

APPLIED PHYSICS LETTERS

Kim, Minki; Goh, Youngin; Hwang, Junghyeon; Jeon, Sanghun

被引用次数: 13

2021

Effect of Ga composition on mobility in a-InGaZnO thin-film transistors

NANOTECHNOLOGY

Ahn, Minho; Gaddam, Venkateswarlu; Park, Sungho; Jeon, Sanghun

被引用次数: 2

2021

High-Performance and High-Endurance HfO₂-Based Ferroelectric Field-Effect Transistor Memory with a Spherical Recess Channel

PHYSICA STATUS SOLIDI-RAPID RESEARCH LETTERS

Kim, Taeho; Hwang, Junghyeon; Kim, Giuk; Jung, Minhyun; Jeon, Sanghun

被引用次数: 20

2021

Insertion of Dielectric Interlayer: A New Approach to Enhance Energy Storage in Hf_xZr_{1-x}O₂ Capacitors

IEEE ELECTRON DEVICE LETTERS

Das, Dipjyoti; Gaddam, Venkateswarlu; Jeon, Sanghun

被引用次数: 35

2021

Flexible multimodal sensor inspired by human skin based on hair-type flow, temperature, and pressure

FLEXIBLE AND PRINTED ELECTRONICS

Jung, Minhyun; Lee, Jumi; Vishwanath, Sujaya Kumar; Kwon, Oh-Sun; Ahn, Chi Won; Shin, Kwanwoo; Jeon, Sanghun

被引用次数: 26

2020

Crystalline Phase-Controlled High-Quality Hafnia Ferroelectric With RuO₂ Electrode

IEEE TRANSACTIONS ON ELECTRON DEVICES

Goh, Youngin; Cho, Sung Hyun; Park, Sang-Hee Ko; Jeon, Sanghun

被引用次数: 43

2020

High-k Hf_xZr_{1-x}O₂ Ferroelectric Insulator by Utilizing High Pressure Anneal

IEEE TRANSACTIONS ON ELECTRON DEVICES

Das, Dipjyoti; Jeon, Sanghun

被引用次数: 48

2020

Insertion of HfO₂ Seed/Dielectric Layer to the Ferroelectric HZO Films for Heightened Remanent Polarization in MFM Capacitors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Gaddam, Venkateswarlu; Das, Dipjyoti; Jeon, Sanghun

被引用次数: 91

2020

Oxygen vacancy control as a strategy to achieve highly reliable hafnia ferroelectrics using oxide electrode

NANOSCALE

Goh, Youngin; Cho, Sung Hyun; Park, Sang-Hee Ko; Jeon, Sanghun

被引用次数: 122

2020

Effect of Forming Gas High-Pressure Annealing on Metal-Ferroelectric-Semiconductor Hafnia Ferroelectric Tunnel Junction

IEEE ELECTRON DEVICE LETTERS

Hwang, Junghyeon; Goh, Youngin; Jeon, Sanghun

被引用次数: 55

2020

Evolution of crystallographic structure and ferroelectricity of Hf_{0.5}Zr_{0.5}O₂ films with different deposition rate

AIP ADVANCES

Kim, Taeho; An, Minho; Jeon, Sanghun

被引用次数: 25

2020

Excellent Reliability and High-Speed Antiferroelectric HfZrO₂ Tunnel Junction by a High-Pressure Annealing Process and Built-In Bias Engineering

ACS APPLIED MATERIALS & INTERFACES

Goh, Youngin; Hwang, Junghyeon; Jeon, Sanghun

被引用次数: 49

2020

Metal-induced n⁺/n homojunction for ultrahigh electron mobility transistors

NPG ASIA MATERIALS

Park, Ji-Min; Kim, Hyoung-Do; Joh, Hongrae; Jang, Seong Cheol; Park, Kyung; Park, Yun Chang; Nahm, Ho-Hyun; Kim, Yong-Hyun; Jeon, Sanghun; Kim, Hyun-Suk

被引用次数: 12

2020

Metallic Glass-Based Dual-Mode Sensor for Soft Electronics

IEEE SENSORS JOURNAL

Jung, Minhyun; Yun, Changjin; Kim, Mingu; Rhie, Kungwon; Jeon, Sanghun

被引用次数: 2

2020

Trade-off between interfacial charge and negative capacitance effects in the Hf-Zr-Al-O/Hf_{0.5}Zr_{0.5}O₂ bilayer system

SOLID-STATE ELECTRONICS

Das, Dipjyoti; Kim, Taeho; Gaddam, Venkateswarlu; Shin, Changhwan; Jeon, Sanghun

被引用次数: 17

2020

Broad spectral responsivity in highly photoconductive InZnO/MoS₂ heterojunction phototransistor with ultrathin transparent metal electrode

NANOTECHNOLOGY

Das, Dipjyoti; Park, Junghak; Ahn, Minho; Park, Sungho; Hur, Jihyun; Jeon, Sanghun

被引用次数: 0

2020

Ultra-thin Hf_{0.5}Zr_{0.5}O₂ thin-film-based ferroelectric tunnel junction via stress induced crystallization

APPLIED PHYSICS LETTERS

Goh, Youngin; Hwang, Junghyeon; Lee, Yongsun; Kim, Minki; Jeon, Sanghun

被引用次数: 111

2020

Demonstration of High Ferroelectricity ($P_r \sim 29 \mu\text{C}/\text{cm}^2$) in Zr Rich Hf_xZr_{1-x}O₂ Films

IEEE ELECTRON DEVICE LETTERS

Das, Dipjyoti; Gaddam, Venkateswarlu; Jeon, Sanghun

被引用次数: 44

2020

Robust and scalable three-dimensional spacer textile pressure sensor for human motion detection

SMART MATERIALS AND STRUCTURES

Kim, Kyungkwan; Jung, Minhyun; Jeon, Sanghun; Bae, Jihyun

被引用次数: 40

2019

Comprehensive study of high pressure annealing on the ferroelectric properties of Hf_{0.5}Zr_{0.5}O₂ thin films

NANOTECHNOLOGY

Oh, Changyong; Tewari, Amit; Kim, Kyungkwan; Kumar, Ulayil Sajesh; Shin, Changhwan; Ahn, Minho; Jeon, Sanghun

被引用次数: 8

2019

Improved optical performance of multi-layer MoS₂ phototransistor with see-through metal electrode

NANO CONVERGENCE

Park, Junghak; Das, Dipjyoti; Ahn, Minho; Park, Sungho; Hur, Jihyun; Jeon, Sanghun

被引用次数: 11

2019

Flexible Multimodal Sensors for Electronic Skin: Principle, Materials, Device, Array Architecture, and Data Acquisition Method

PROCEEDINGS OF THE IEEE

Jeon, Sanghun; Lim, Soo-Chul; Tran Quang Trung; Jung, Minhyun; Lee, Nae-Eung

被引用次数: 70

2019

Influence of High-Pressure Annealing on Memory Properties of Hf_{0.5}Zr_{0.5}O₂ Based 1T-FeRAM

IEEE ELECTRON DEVICE LETTERS

Yoon, Jae Seok; Tewari, Amit; Shin, Changhwan; Jeon, Sanghun

被引用次数: 13

2019

Amorphous FeZr metal for multi-functional sensor in electronic skin

NPJ FLEXIBLE ELECTRONICS

Jung, Minhyun; Lee, Eunha; Kim, Dongseuk; Kim, Kyungkwan; Yun, Changjin; Lee, Hyangsook; Kim, Heegoo; Rhie, Kungwon; Jeon, Sanghun

被引用次数: 20

2019

Transparent and Flexible Mayan-Pyramid-based Pressure Sensor using Facile-Transferred Indium tin Oxide for Bimodal Sensor Applications

SCIENTIFIC REPORTS

Jung, Minhyun; Vishwanath, Sujaya Kumar; Kim, Jihoon; Ko, Dae-Kwan; Park, Myung-Jin; Lim, Soo-Chul; Jeon, Sanghun

被引用次数: 26

2019

Effects of high pressure nitrogen annealing on ferroelectric Hf_{0.5}Zr_{0.5}O₂ films

APPLIED PHYSICS LETTERS

Kim, Taeho; Park, Jinsung; Cheong, Byoung-Ho; Jeon, Sanghun

被引用次数: 58

2018

Non-volatile resistive switching in CuBi-based conductive bridge random access memory device

APPLIED PHYSICS LETTERS

Vishwanath, Sujaya Kumar; Woo, Hyunsuk; Jeon, Sanghun

被引用次数: 9

2018

Enhanced tunneling electroresistance effects in HfZrO-based ferroelectric tunnel junctions by high-pressure nitrogen annealing

APPLIED PHYSICS LETTERS

Goh, Youngin; Jeon, Sanghun

被引用次数: 62

2018

Pulse Switching Study on the HfZrO Ferroelectric Films With High Pressure Annealing

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Taeho; Jeon, Sanghun

被引用次数: 55

2018

First-order reversal curve diagrams for characterizing ferroelectricity of Hf_{0.5}Zr_{0.5}O₂ films grown at different rates

JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B

Goh, Youngin; Jeon, Sanghun

被引用次数: 15

2018

Excellent Resistive Switching Performance of Cu-Se-Based Atomic Switch Using Lanthanide Metal Nanolayer at the Cu-Se/Al₂O₃ Interface

ACS APPLIED MATERIALS & INTERFACES

Woo, Hyunsuk; Vishwanath, Sujaya Kumar; Jeon, Sanghun

被引用次数: 15

2018

Discharge Current Analysis Estimating the Defect Sites in Amorphous Hafnia Thin-Film Transistor

IEEE TRANSACTIONS ON ELECTRON DEVICES

Goh, Youngin; Jeon, Sanghun

被引用次数: 0

2018

Resistive switching characteristics of a modified active electrode and Ti buffer layer in Cu-Se-based atomic switch

JOURNAL OF ALLOYS AND COMPOUNDS

Woo, Hyunsuk; Vishwanath, Sujaya Kumar; Jeon, Sanghun

被引用次数: 16

2018

Effect of dysprosium and lutetium metal buffer layers on the resistive switching characteristics of Cu-Sn alloy-based conductive-bridge random access memory

NANOTECHNOLOGY

Vishwanath, Sujaya Kumar; Woo, Hyunsuk; Jeon, Sanghun

被引用次数: 7

2018

Enhancement of resistive switching properties in Al₂O₃ bilayer-based atomic switches: multilevel resistive switching

NANOTECHNOLOGY

Vishwanath, Sujaya Kumar; Woo, Hyunsuk; Jeon, Sanghun

被引用次数: 58

2018

Quantitative analysis of charge trapping and classification of sub-gap states in MoS₂ TFT by pulse I-V method

NANOTECHNOLOGY

Park, Junghak; Hur, Ji-Hyun; Jeon, Sanghun

被引用次数: 13

2018

Scalable and facile synthesis of stretchable thermoelectric fabric for wearable self-powered temperature sensors

RSC ADVANCES

Jung, Minhyun; Jeon, Sanghun; Bae, Jihyun

被引用次数: 66

2018

Transparent Semiconducting Oxide Technology for Touch Free Interactive Flexible Displays

PROCEEDINGS OF THE IEEE

Lee, Sungsik; Jeon, Sanghun; Chaji, Reza; Nathan, Arokia

被引用次数: 92

2015

Origin of High Photoconductive Gain in Fully Transparent Heterojunction Nanocrystalline Oxide Image Sensors and Interconnects

ADVANCED MATERIALS

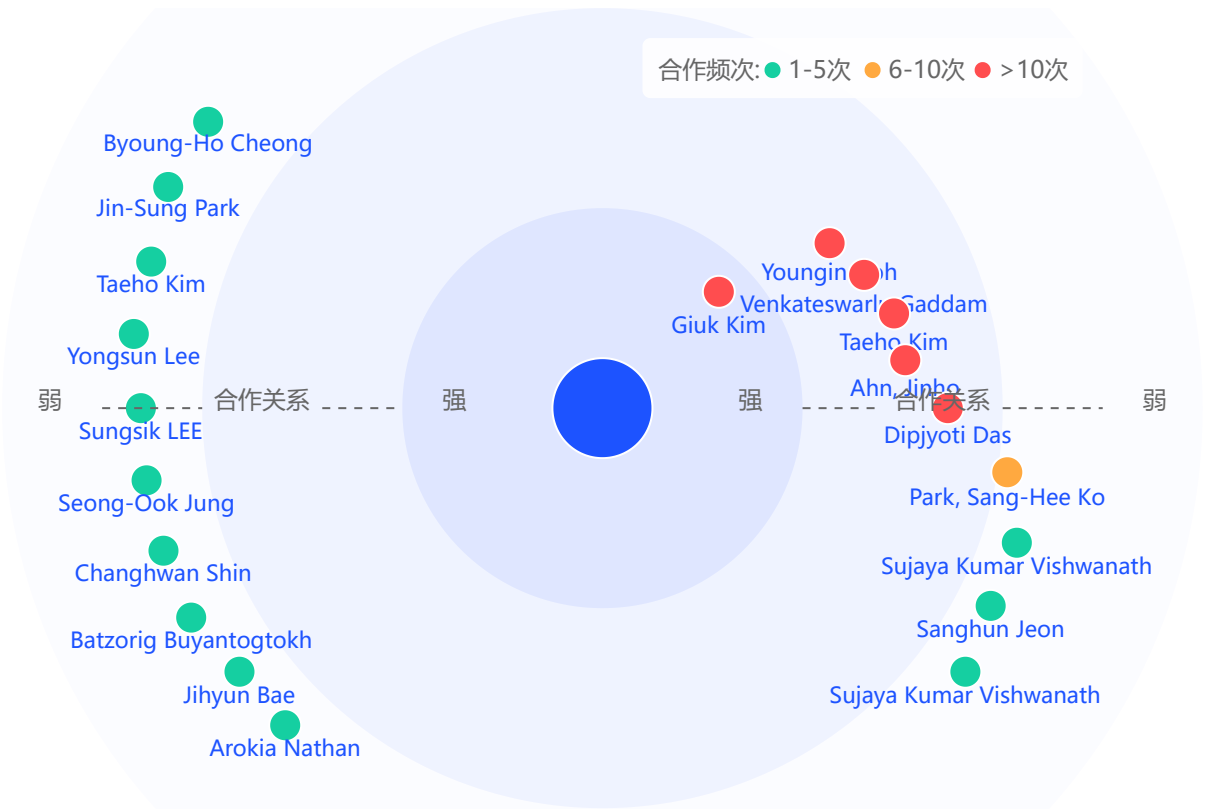
Transparent Semiconducting Oxide Technology for Touch Free Interactive Flexible Displays

Proc.IEEE

Sungsik Lee; Sanghun Jeon; Reza Chaji; Arokia Nathan

被引用次数：0

学术合作



Giuk Kim	合作次数：26
韩国科学技术院	
Youngin Goh	合作次数：17
韩国科学技术院	
Venkateswarlu Gaddam	合作次数：16
佛罗里达大学	
Taeho Kim	合作次数：15
韩国科学技术院	
Ahn, Jinho	合作次数：15
汉阳大学	

还有 26 个合作者